Weekly Student Progress Performa

Team Id	CS26084	**Week no / Date**	Week 10 21 September 2026
Project Title	SentinelCore: A Network Detection and Incident Response Platform		
Members Present	Om Rashiya (24cs084), Yash Patel (24cs082)		
Mentor	Akshita Kadam		

WEEK 10 • Search, Correlation, Incidents, and Reporting Start

Section	Details to be filled by the team
Planned Tasks (from last week)	Complete search, correlation, and incident management; start evidence reporting and response actions.
Tasks Completed This Week	Completed event search with facets/saved searches, four correlation evaluator types and live worker execution, and incident lifecycle management. Defined asynchronous report generation and a protected firewall response model with expiry and recovery behavior.
Commits Made	Total: 3 feature commits. 567e05d — M6 event search; 76d671a — M7 correlation engine; ba4c083 — M8 incident management.
Issues Created / Closed	Created: Long report jobs should not block API requests; containment must never block protected addresses. Closed: Designed queued report generation and mandatory protection/TTL checks before firewall changes.
Milestone Status	~82% complete — on track. Detection-to-incident workflow was complete and response/reporting work had begun.
Blockers / Challenges Faced	Keeping evaluator queries bounded and ensuring incident promotion remained idempotent under repeated worker execution.
Plan for Next Week	Finish PDF/CSV/JSON reports, schedules, firewall containment, PCAP upload/analysis, and threat intelligence.

Research and Technical Study

Research objective. Complete search, correlation, and incident management; start evidence reporting and response actions.

Study completed. Studied asynchronous report pipelines, reproducible evidence exports, iptables containment, protected-IP guardrails, TTL-based rollback, and crash reconciliation. Response actions require explicit user intent and retain a complete audit trail.

Report types and output mapping

Report type	Primary content	Supported project output
Incident summary	Incident queue and totals	PDF / CSV / JSON
Incident detail	Narrative and linked evidence	PDF / JSON
Asset inventory	Assets and discovered services	PDF / CSV / JSON
Event statistics	Counts and grouped event data	PDF / CSV / JSON
Scheduling	Stored report parameters	Queued generation by the worker

Decision carried into the project

Completed event search with facets/saved searches, four correlation evaluator types and live worker execution, and incident lifecycle management. Defined asynchronous report generation and a protected firewall response model with expiry and recovery behavior.

Implementation Record and Website Evidence

M6 completion

Delivered keyset-paginated event search, bounded filters, facets, saved searches, and frontend query-state handling in commit 567e05d.

M7 completion

Delivered stored, explainable correlation rules with four evaluator types, candidate generation, run tracking, and live worker execution in commit 76d671a.

M8 completion

Delivered promotion, queue management, assignment, lifecycle transitions, evidence handling, comments, and history in commit ba4c083.

Reporting design

Defined four report types, asynchronous generation, PDF/CSV/JSON output where applicable, report status/download handling, and administrator-managed schedules without blocking interactive API calls.

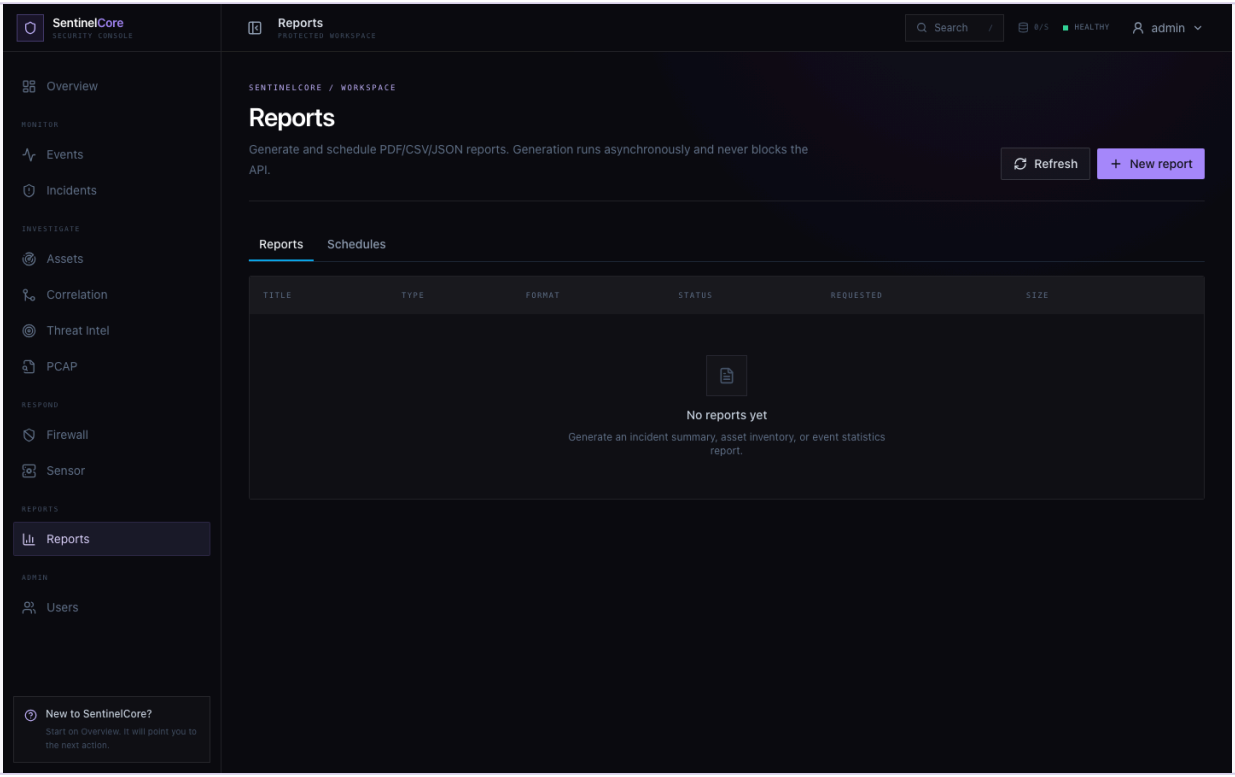


Figure W10.1 — Project-owned SentinelCore interface mapped specifically to Week 10.

Evidence interpretation. The Reports workspace provides Reports and Schedules tabs, refresh and New report actions, columns for title/type/format/status/requested/size, and report-oriented empty-state guidance. Its visible controls, labels, status fields, and navigation provide direct evidence for the work described on this page.

Repository and Architecture Evidence

Asynchronous report generator

backend/app/reports/generator.py

```
43      return _storage_root() / f"{report_id}.{_EXT[fmt]}"
44
45
46  async def generate_report(report_id: uuid.UUID, sessionmaker: async_sessionmaker[AsyncSession]) -> None:
47      async with sessionmaker() as db:
48          report = await db.get(Report, report_id)
49          if report is None or report.status != ReportStatus.QUEUED:
50              return
51          report.status = ReportStatus.RUNNING
52          report.started_at = datetime.now(timezone.utc)
53          await db.commit()
54
55      try:
56          async with sessionmaker() as db:
57              report = await db.get(Report, report_id)
58              builder = REGISTRY[report.report_type]
59              data = await asyncio.wait_for(
60                  builder.build(report.params, db), timeout=settings.report_generation_timeout_seconds
61              )
62
63              requester = await db.get(User, report.requested_by) if report.requested_by else None
64              requested_by_display = requester.username if requester else "system"
65
66              if report.format == ReportFormat.PDF:
```

Repository evidence captured directly from the SentinelCore workspace

Figure W10.2 — Asynchronous report generator. This is a separate internal artifact, not a copy of the website image.

Repository files used for traceability

- reports/modules/M06-search_1.md
- reports/modules/M07-correlation_1.md
- reports/modules/M08-incidents_1.md
- reports/modules/M09-reporting_1.md
- frontend/src/pages/reports/ReportsPage.jsx

Verification

The three feature commits and reporting controls were taken directly from Git history and implemented module files.

Traceability conclusion. Asynchronous report generator directly supports the Search, Correlation, Incidents, and Reporting Start milestone. The three feature commits and reporting controls were taken directly from Git history and implemented module files. The artifact, source paths, implementation record, and website evidence therefore describe the same weekly scope.

Weekly Closure and Faculty Assessment

Completed outcome

Completed event search with facets/saved searches, four correlation evaluator types and live worker execution, and incident lifecycle management. Defined asynchronous report generation and a protected firewall response model with expiry and recovery behavior.

Commit record

Total: 3 feature commits. 567e05d — M6 event search; 76d671a — M7 correlation engine; ba4c083 — M8 incident management.

Issue and resolution

Created: Long report jobs should not block API requests; containment must never block protected addresses. Closed: Designed queued report generation and mandatory protection/TTL checks before firewall changes.

Milestone position

~82% complete — on track. Detection-to-incident workflow was complete and response/reporting work had begun.

Challenge handled

Keeping evaluator queries bounded and ensuring incident promotion remained idempotent under repeated worker execution.

Next-week hand-off

Finish PDF/CSV/JSON reports, schedules, firewall containment, PCAP upload/analysis, and threat intelligence.

Repository reference

https://github.com/rashiyaom/SentinelCore

Assessed By (Faculty)	Marks Awarded (5 marks)	Faculty Remarks